

TEO.: SEAN $G = \langle a \rangle$ y G' GPOS

1) SI $o(a) = +\infty$; PARA CADA $b \in G'$
 $\exists! f \in \text{Hom}(G; G')$ TAL QUE
 $f(a) = b$

2) SI $o(a) < +\infty$; PARA CADA $b \in G'$
 CON $o(b) | o(a)$; $\exists! f \in \text{Hom}(G; G')$
 TAL QUE $f(a) = b$

P.: DEFINO $f: G \rightarrow G'$, $\forall k \in \mathbb{Z}$
 $a^k \mapsto b^k$

* SI $o(a) = +\infty$; $\forall z \in G$; $\exists! k \in \mathbb{Z}$ /
 $z = a^k$

* BUENA DEF. (CASO INF.)

SI $a^k = a^r \rightarrow a^{k-r} = e$
 SUPONG. QUE $k-r \neq 0$
 $\rightarrow o(a) | (k-r)$
 $(\rightarrow \leftarrow)$ YA QUE $o(a) = +\infty$
 $\therefore k=r$

* $f \in \text{Hom}(G; G')$

$f(a^k \cdot a^r) = f(a^{k+r}) = b^{k+r}$
 $= b^k \cdot b^r$
 $= f(a^k) \cdot f(a^r)$

* BUENA DEF. (CASO FINITO)

SI $a^k = a^r \rightarrow a^{k-r} = e$
 $\rightarrow o(a) | (k-r)$
 $\xrightarrow{\text{TRANS}} o(b) | (k-r)$
 $\rightarrow b^{k-r} = e$
 $\rightarrow \underbrace{b^k}_{f(a^k)} = \underbrace{b^r}_{f(a^r)}$

* UNICIDAD.

SUPONG. QUE $\exists g \in \text{Hom}(G; G')$ /
 $g(a^k) = b^k = f(a^k)$, $\forall k \in \mathbb{Z}$
 $\rightarrow f = g$

EJM.: DETERMINE

$\text{Hom}(\mathbb{Z}_8; \mathbb{Z}_{10})$
 Como $\mathbb{Z}_8 = \langle \bar{1} \rangle$ y $o(\bar{1}) = 8$

$\xrightarrow[\text{ANT.}]{\text{XTEO}}$ PARA CADA $b \in \mathbb{Z}_{10}$
 CON $o(b) | 8$

$\exists! f \in \text{Hom}(\mathbb{Z}_8; \mathbb{Z}_{10})$

* $o(b) = 1$: $\underbrace{b = \bar{0}}$ * $o(b) = 2$: $\underbrace{b = \bar{5}}$

$f_1(\bar{1}) = \bar{0}$ $f_2(\bar{1}) = \bar{5}$ $\rightarrow \text{Hom}(\mathbb{Z}_8; \mathbb{Z}_{10}) = \{f_1, f_2\}$
 $f_1(\bar{m}) = \bar{0}$ $f_2(\bar{m}) = \bar{5m}$

TEO.: PARA CADA $n \in \mathbb{Z}_{\geq 2}$
 SE CUMPLE

$\text{Aut}(\mathbb{Z}_n) \simeq \mathcal{U}(\mathbb{Z}_n)$

P.: DEFINO

$\psi: \text{Aut}(\mathbb{Z}_n) \rightarrow \mathcal{U}(\mathbb{Z}_n)$
 $f \mapsto f(\bar{1})$

* ψ ES HOMOM. DE GPOS.

$\psi(f \circ g) = (f \circ g)(\bar{1})$

$= f(g(\bar{1}))$

$= f(\psi(g))$

$= \psi(f) \psi(g)$

TEO.: PARA CADA $n \in \mathbb{Z}_{\geq 2}$
SE CUMPLE

$$\text{Aut}(\mathbb{Z}_n) \simeq \mathcal{U}(\mathbb{Z}_n)$$

P.: DEFINO

$$\psi: \text{Aut}(\mathbb{Z}_n) \rightarrow \mathcal{U}(\mathbb{Z}_n)$$

$$f \mapsto f(1)$$

* ψ ES HOMOM. DE GRP.

$$\psi(f \circ g) = (f \circ g)(1)$$

$$= f(g(1))$$

$$= f(\psi(g))$$

$$= \psi(f) \circ \psi(g)$$

$$\mathbb{Z}_0 = \{f_1, f_2\}$$

* ψ ES INYECT.

$$\text{SI } \psi(f) = \bar{1} \rightarrow f(1) = \bar{1}$$

$$\rightarrow f(\bar{m}) = \bar{m}$$

$$\rightarrow f = \text{id}_{\mathbb{Z}_n}$$

* ψ ES SOBREV.

$$\text{SEA } \bar{m} \in \mathcal{U}(\mathbb{Z}_n) \rightarrow (m, n) = 1$$

$$\text{XBEZANT} \rightarrow \exists n, r, s \in \mathbb{Z} /$$

$$rm + sn = 1$$

$$\text{DEFINO } f_m(\bar{r}) = \overline{rm}$$

$$\bullet f_m(\bar{a} + \bar{b}) = f_m(\overline{a+b}) = \overline{(a+b)m}$$

$$= \overline{am + bm} = f_m(\bar{a}) + f_m(\bar{b})$$

$$\bullet \text{SI } f_m(\bar{r}) = \bar{0}$$

$$\rightarrow \overline{km} = \bar{0}$$

$$\rightarrow n | km$$

$$\rightarrow n | k \rightarrow \bar{k} = \bar{0}$$

$$\rightarrow f_m \text{ ES INV.}$$

$$\bullet f_m \text{ ES SOBREV.}$$

$$\text{SEA } \bar{x} \in \mathbb{Z}_n \text{ CUALQ.}$$

$$\rightarrow \bar{x} = \overline{x \cdot 1}$$

$$= \overline{x \cdot (rm + sn)}$$

$$= \overline{xrm}$$

$$= f_m(\bar{xr})$$

* ψ ES INYECT.

$$\text{SI } \psi(f) = \bar{1} \rightarrow f(\bar{1}) = \bar{1}$$

$$\rightarrow f(\bar{m}) = \bar{m}$$

$$\rightarrow f = \text{id}_{\mathbb{Z}_n}$$

* ψ ES SOBREV.

$$\text{SEA } \bar{m} \in U(\mathbb{Z}_n) \rightarrow (m, n) = 1$$

$$\times \text{BEZOUT} \rightarrow \exists r, s \in \mathbb{Z} /$$

$$rm + sn = 1$$

$$\text{DEFINIR } f_m(\bar{r}) = \overline{rm}$$

$$\begin{aligned} f_m(\overline{a+b}) &= f_m(\overline{a+b}) = \overline{(a+b)m} \\ &= \overline{am + bm} = f_m(\bar{a}) + f_m(\bar{b}) \end{aligned}$$

$$\bullet \text{ SI } f_m(\bar{r}) = \bar{0}$$

$$\rightarrow \overline{km} = \bar{0}$$

$$\rightarrow n | km$$

$$\rightarrow n | k \rightarrow \bar{k} = \bar{0}$$

$$\rightarrow f_m \text{ ES INY.}$$

$$\bullet f_m \text{ ES SOBREV.}$$

$$\text{SEA } \bar{x} \in \mathbb{Z}_n \text{ CUALQ.}$$

$$\begin{aligned} \rightarrow \bar{x} &= \overline{x \cdot 1} \\ &= \overline{x \cdot (rm + sn)} \\ &= \overline{xrm} \\ &= f_m(\bar{x}r) \end{aligned}$$

EJM.: PARA $n=4$

$$U(\mathbb{Z}_4) = \{\bar{1}, \bar{3}\}$$

$$\text{Aut}(\mathbb{Z}_4) = \{f_1, f_2\}$$

$$f_1: \mathbb{Z}_4 \rightarrow \mathbb{Z}_4, f_2: \mathbb{Z}_4 \rightarrow \mathbb{Z}_4$$

$$\bar{1} \mapsto \bar{1}$$

$$\bar{2} \mapsto \bar{2}$$

$$\bar{3} \mapsto \bar{3}$$

$$\bar{0} \mapsto \bar{0}$$

$$\bar{1} \mapsto \bar{3}$$

$$\bar{2} \mapsto \bar{2}$$

$$\bar{3} \mapsto \bar{1}$$

$$\bar{0} \mapsto \bar{0}$$

$$(\bar{1}, -)$$

$$(\bar{2}, -)$$

$$(\bar{3}, -)$$

DEF.: SEA G GPO Y $S \subseteq G$

DEFINAMOS EL GRATO DE CAYLEY

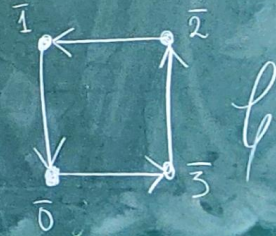
$\mathcal{C} = (V, E)$ COMO EL GRATO CON

$$V = G \wedge E = \{(g, h) / \exists s \in S$$

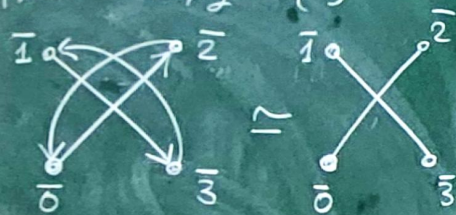
$$\text{TAL QUE } gs = h\}$$

$$= \{(g, h) / g^{-1}h \in S\}$$

EJM.: SEA $G = (\mathbb{Z}_4, +)$ Y $S = \{\bar{3}\}$



2) PARA $G = \mathbb{Z}_4$ y $S = \{\bar{2}\} = S^{-1}$



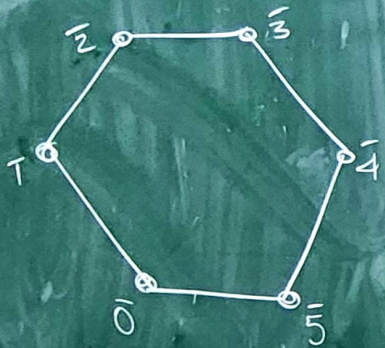
PROP.: SEA G UN GPO Y $S \subseteq G$
 TAL QUE $S = S^{-1}$

→ EL GRAFO DE CAYLEY $\mathcal{C}$ ES
 GRAFO NO DIRG.

P.: EJERC.

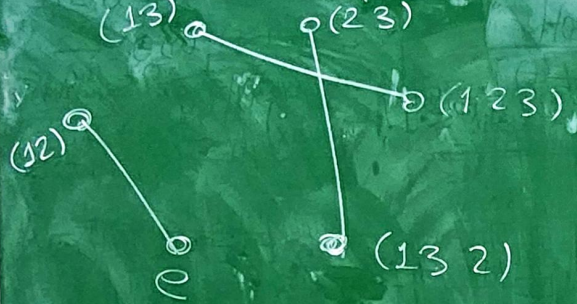
EJERC. 1) SEA $G = \mathbb{Z}_6$

y $S = \{\bar{1}, \bar{5}\} = S^{-1}$



OBS.: DADO G GPO CON $S \subseteq G$
 Y $S = S^{-1}$ TAL QUE $\exists$ XES
 GENERADOR DE $G \rightarrow$ PES CONEXO.

2) SEA $G = S_3$ y $S = \{(1\ 2)\} = S^{-1}$



$(1\ 3)(2\ 3) = (1\ 3\ 2)$	$(2\ 3)(1\ 2) = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \neq (1\ 2)$
$(1\ 3)(1\ 2\ 3) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = (1\ 2)$	$(2\ 3)(1\ 3) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \neq (1\ 2)$
$(1\ 3)(1\ 3\ 2) = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \neq (1\ 2)$	$(2\ 3)(1\ 2\ 3) = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \neq (1\ 2)$
	$(2\ 3)(1\ 3\ 2) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = (1\ 2)$

$\langle \bar{1} \rangle$ y $O(\bar{1}) = 8$
 24 $b \in \mathbb{Z}_{40}$